

Xuming Huang

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Systems for AI; Edge Inference; Operating Systems; AI Compilers

Research Interests

My research focuses on efficient AI inference on heterogeneous edge devices, with an emphasis on compiler optimizations, memory management, and runtime scheduling.

Education

University of Wisconsin–Madison

B.S. in Computer Sciences

GPA: **3.96**/4.00

Madison, WI

Jan. 2025 – May 2027 (Expected)

Stanford University

Summer Session, Visiting Undergraduate

GPA: **4.00**/4.00

Stanford, CA

Jun. 2025 – Aug. 2025

University of Shanghai for Science and Technology

Undergraduate study in Computer Science and Technology

GPA: **3.82**/4.50

Shanghai, China

Sept. 2021 – Jul. 2024

Research Experience

Edge AI Systems Optimization, WukLab, UC San Diego

Jan. 2026 – Present

Undergraduate Researcher advised by Prof. Yiyang Zhang

- Developed a JIT (just-in-time) NPU runtime system that parallelizes progressive graph compilation on CPU with LLM inference on NPU; reduced foreground compilation time by 91.98%, saved 33.05% averaged KV memory
- Optimized the OpenVINO NPU compiler by caching sorted dependency and consumer vectors; reduced CPU sorting time by 94.2%, and contributed [PR #348](#) to the Intel NPU Compiler repository
- Developed a CPU/GPU/NPU benchmarking methodology to characterize operator and inference-stage affinity across edge devices; identified hardware-dependent placement preferences to guide heterogeneous scheduling

LinuxGuard: AI for System Security, UW–Madison [\[Page\]](#)

Jan. 2025 – Nov. 2025

Undergraduate Researcher advised by Prof. Remzi Arpaci-Dusseau and Dr. Vinay Banakar

- Built an LLM-driven pipeline over 50K+ Linux kernel commits to extract bug patterns and synthesize Clang-Tidy analyzers; reduced manual vulnerability triage from weeks to hours
- Developed a multi-stage checker synthesis pipeline with K-means clustering over 10K+ code patterns and compiler-feedback repair; increased valid checkers 4x while achieving 73% precision
- Engineered a full-kernel scanning and validation harness to evaluate checker failures and false positives; uncovered 43 long-latent kernel bugs with an average age of 4.7 years

Operating Systems for AI Inference, UW Madison [\[Page\]](#)

Jan. 2026 – May 2026

Honors Course Research Project advised by Prof. Michael Swift

- Evaluated filesystem interference on LLM inference using controlled background file reads and fixed inference workloads; measured 17.1% higher median latency than baseline
- Investigated inference slowdown using Windows PerfMon; observed stable working sets and page-fault counts plateauing near 29.5K despite rising latency, supporting shared-resource contention over working-set thrashing

ML-Guided Scheduling for Heterogeneous Compute Systems, USST [\[Paper\]](#)

Jan. 2024 – Sept. 2026

Coauthor | Undergraduate Researcher advised by Prof. Xing Hu

- Evaluated an ML-guided scheduler that predicts workload–hardware affinity using calibrated DCN/XGBoost classifiers and places tasks across heterogeneous CPU/GPU pools based on runtime, energy, and resource availability
- Contributed to experimental validation and performance analysis on MIT Supercloud traces, achieving 97.1% resource-matching accuracy, 46.5% lower response time than Meta-RHDC in 5,000-task simulations

Publications

[3] CASH: Confidence-Calibrated Deep Feature Crossing for Classification-Aware Heterogeneous Task Scheduling.

Chuanlin Jian, Yuanchen Sun, Xiangcheng Liu, Xuming Huang, Samaneh Beheshti Kashi, and Xing Hu
Computers, Materials & Continua, 89(2), Article 29, 2026. [DOI](#)

[2] A Method for Laryngoscopic Image Classification: Leveraging Self-supervised BEIT-style Pretraining.

Jie Kong, Haotian Wu, Xuming Huang, Liangjie Wu, and Xing Hu
In *7th International Conference on Frontier Technologies of Information and Computer*, pp. 701–706, 2025. [DOI](#)

[1] Near-infrared tuneable filter based on polymer dispersed liquid crystals.

Ruimei Zeng, Rongbo Bi, Junyang Zhu, Xuming Huang, and Qi Wang
Micro & Nano Letters, 20(1), e70007, 2025. [DOI](#)

Teaching Experience

ECE 252, Department of Electrical and Computer Engineering, UW–Madison
ICPC (In-Class Peer Coach)

Sept. 2026 - Dec. 2026

- Support discussion session and hold office hours for ECE 252

Selected Projects

Open-source Learning Resources and Academic Website [[Algo](#), [Transformer](#), [Webpage](#)]
Open-source Author and Maintainer

Jan. 2024 – Present

- Developed and maintained 20+ algorithm and data-structure implementations across 6+ topics, with visualizations and practice templates; reached 100+ GitHub stars
- Implemented Transformer and GPT-2 from scratch in PyTorch, covering multi-head attention, encoder-decoder architecture, optimization, and training
- Built and open-sourced a personal academic website and reusable portfolio template; reached 30+ GitHub stars

Heap Allocator

Systems Project during Stanford Summer Session

Jun. 2025 – Aug. 2025

- Implemented free-block management, coalescing, and boundary tags; explored fragmentation and allocator behavior through randomized stress tests and memory-debugging tools

Work Experience

Cool AI Technology

Technical R&D and Product Development Intern, Shanghai

Jul. 2024 – Sept. 2024

- Redesigned the company’s web interface using Next.js and Tailwind CSS, improving page load time by 45% and increasing user engagement metrics by 30% across 5000+ daily active users
- Architected a scalable FastAPI backend for LLM integration, handling 1,000+ concurrent requests and reducing API response time from 3s to 800ms
- Deployed Prompted Agents for AI-Hub platform, serving 20+ enterprise clients, generating \$10K in new revenue within the first month of launch

Technical Skills

Languages: Python, C/C++, Java, JavaScript, Verilog, assembly, Swift

Systems and tools: Linux, OpenVINO, LLVM/Clang, Git, Docker, GDB, Valgrind

Methods: Performance measurement, compiler/runtime experiments, model fine-tuning, correctness evaluation

Honors and Awards

- Dean’s List, University of Wisconsin–Madison
- Presidential Scholarship, University of Shanghai for Science and Technology (Top 1%)